

Funding Proposal: GPU Compute Node for the Campus Research Cluster

Prepared by Jeremy R. Manning | June 2026 | Request: **\$185,000** (within \$150K–\$200K envelope) | One-time capital, 3-year warranty included

Summary

We propose a one-time investment of **\$185,000** to purchase a flagship GPU compute node and integrate it into the existing campus research cluster (Slurm-scheduled, InfiniBand-connected). The node — four NVIDIA H200 accelerators with 564 GB of pooled high-bandwidth memory — closes a critical gap in large-memory accelerated computing that currently forces our LLM development and automated-discovery work onto metered cloud GPUs. As a shared cluster resource it will serve multiple labs while directly accelerating model fine-tuning, high-throughput inference, and the autonomous research pipeline.

Justification

- **A real capability gap.** Training, fine-tuning, and serving modern open models (70B–400B parameters) require 100+ GB of GPU memory and NVLink-coupled accelerators. The campus cluster currently lacks large-VRAM GPUs, so this work either stalls in queue or is pushed to costly on-demand cloud instances.
- **Cost-effective vs. cloud.** A comparable 4×H200 cloud instance rents for ~\$14–16/hr; at sustained research utilization that is ~\$90K–\$120K per year. The node pays for itself in roughly 18–24 months and then delivers 2–3 further years of compute at only power and maintenance cost — on-premises, with no data leaving campus.
- **Enables automated scientific discovery.** Our llmXive platform runs ~50 specialist LLM agents that brainstorm, draft, and peer-review research artifacts. Local inference lets these pipelines run continuously without per-token API fees (advancing our free-first cost principle) and unlocks fine-tuning of domain-specialist models.
- **Shared, well-utilized asset.** The node joins the existing Slurm scheduler, so idle cycles are immediately available to other groups across the institution — maximizing return on a single capital purchase.

Proposed System & Budget

Component	Specification	Cost (USD)
GPU accelerators	4 × NVIDIA H200 SXM, 141 GB HBM3e (HGX baseboard, NVLink/NVSwitch); 564 GB pooled	\$128,000
Server platform	Dual AMD EPYC (192 cores), HGX carrier, redundant 3 kW PSUs	\$20,000
System memory	1.5 TB DDR5 ECC	\$10,000
Local storage	~30 TB NVMe (RAID1 OS + high-speed scratch)	\$8,000
Cluster networking	NVIDIA ConnectX-7 NDR 400 Gb InfiniBand HCA + cabling	\$6,000
Integration	Rack/rails/PDU, Slurm integration, burn-in testing	\$3,000
Warranty & support	3-year 24×7 on-site, next-business-day parts	\$10,000
Total		\$185,000

Alternative configuration: for workloads dominated by many-user inference rather than large-model training, the same budget buys 8 × NVIDIA L40S (48 GB) for higher aggregate throughput. The H200 build above is recommended as the more future-proof, training-capable option.

Expected outcomes: continuous on-premises LLM training/inference, elimination of recurring cloud GPU spend at sustained utilization, fine-tuning of specialist research agents, and a shared accelerator resource for the broader campus community. Figures are planning estimates (June 2026) subject to vendor quotes; power and cooling assumed available in the existing cluster facility.